
 $^{36}\text{Ar}(\text{n},\text{d})$ [1977Pa29](#)

$J^\pi=0^+$ for ^{36}Ar ground state.

[1977Pa29](#): 14.1-MeV neutrons were produced via the $^3\text{H}(\text{d},\text{n})^4\text{He}$ reaction at an average rate of 10^9 neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 2.85-mg/cm^2 ^{36}Ar target in the form of a gas cell filled with 99.8% enriched ^{36}Ar . Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}Cl Levels

E(level)	L [†]	Comments
0	2	S: 3.83 48 and 3.90 47 using two sets of optical parameters.
1219	0	S: 1.94 23 and 3.30 40 using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences.
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[†] From DWBA analysis of the measured $\sigma(\theta)$ in [1977Pa29](#).